

MULTIVARIATE CALCULUS - MATH 252

VECTOR FIELDS, LINE, SURFACE AND VOLUME INTEGRAL

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Vector Fields

Definition

A vector field on two (or three) dimensional space is a function $\vec{F}$ that assigns to each point (x,y) (or(x,y,z)) a two (or three) dimensional vector given by $\vec{F}(x,y)$ or $(\vec{F}(x,y,z))$

The standard notation for the function $\vec{F}$ is,
$\vec{F}(x,y) = P(x,y)\vec{i} + Q(x,y)\vec{j}$
$\vec{F}(x,y,z) = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + R(x,y,z)\vec{k}$
The function P, Q, R (if present) are sometimes called scalar functions.

Given a function f(x,y,z) the gradient vector is defined by, $\nabla f = (f_x, f_y, f_z)$. This is a vector field and is often called a gradient vector field. In these cases the function f(x,y,z) is scalar function.

Find the gradient vector field of the following functions:
a. $f(x,y) = x^2 sin(5y)$ b. $f(x,y,z) = ze^{-xy}$

Solution

a. $\nabla f = [2xsin(5y), 5x^2 cos(5y)]$

b. $\nabla f = [-yze^{-xy}, -xze^{-xy}, e^{-xy}]$

A vector field $\vec{F}$ is called a conservative vector field if there exists a function f such that $\vec{F} = \nabla f$. If $\vec{F}$ is a conservative vector field then the function f is called a potential function for $\vec{F}$.

For instance the vector field $\vec{F} = y\vec{i} + x\vec{j}$ is a conservative vector field with a potential function of $f(x,y) = xy$ because $\nabla f = [x,y]$.

On the other hand $\vec{F} = -y\vec{i} + x\vec{j}$ is not a conservative vector field since there is no function f such that $\vec{F} = \nabla f$.



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Line Integrals With Respect to Arc Length

The following are basic curves and their parametric equation:

Curve	Parametric Equations
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$(ellipse)	$x = acos(t), y = bsin(t), 0 \leq t \leq 2\pi$
$x^2 + y^2 = r^2$	$x = rcos(t), y = rsin(t), 0 \leq t \leq 2\pi$
$y = f(x)$	$x = t, y = f(t)$
$x = g(y)$	$x = g(t), y = t$

Curve	Parametric Equations
$\vec{r} = (1-t)\langle x_0, y_0, z_0\rangle + t\langle x_1, y_1, z_1\rangle$	$0 \leq t \leq 1$
Line segment from (x_0, y_0, z_0) to (x_1, y_1, z_1)	$x = (1-t)x_0 + tx_1$ $y = (1-t)y_0 + ty_1, 0 \leq t \leq 1$ $z = (1-t)z_0 + tz_1$

Let the curve C have the following parametric equations x=h(t), y=g(t), $a \leq t \leq b$.
The parameterization of the curve as a vector function is given as:
$\vec{r}(t) = h(t)\vec{i} + g(t)\vec{j}, a \leq t \leq b$
The curve is called smooth if $\vec{r}'(t)$ is continuous.

The line integral of f(x,y) along C is defined by , $\int_C f(x,y)ds$, where ds is the path along the curve C. This is sometimes called the line integral of f with respect to arc length.

Note, the length of a curve given by the parametric equations is: $L = \int_a^b f(x,y)ds$, where $ds = \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}dt$

Converting the line integral $\int_C f(x,y)ds$ into parametric form we have:

$$\int_C f(x,y)ds = \int_a^b f(x(t),y(t))\sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}dt$$

Using the vector form of the parameterization the line integral can be written as:

$\int_C f(x,y)ds = \int_a^b f(h(t),g(t))\|\vec{r}'(t)\|dt$, where $\|\vec{r}'(t)\|$ is the magnitude or norm of $\vec{r}'(t)$.

Note

As long as the parameterization of the curve C is traced out exactly once as t increases from a to b the value of the line integral will be independent of the parameterization of the curve.

E.g. $\int_C xy^4 ds$ where C is the right half of the circle, $x^2 + y^2 = 16$

Solution

The parameterization of the circle is given by, $x = 4cost$, $y = 4sint$

The range of t that gives the right half of the circle is $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$ and

$\frac{dx}{dt} = -4sint$, $\frac{dy}{dt} = 4cost$ and $ds = \sqrt{16sin^2t + 16cos^2t}dt = 4dt$



Then the line integral is

$$\int_C xy^4 ds = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4cost(4sint)^4(4)dt$$

$$= 4096\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} costsin^4 tdt = \frac{4096}{5}sin^5 t\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{8192}{5}$$



Piece-wise smooth curve

A piece-wise smooth curve is any curve that can be written as the union of a finite number of smooth curves, $C_1, \cdots, C_n$ where the end point of C_1 is the starting point of C_{i+1}.

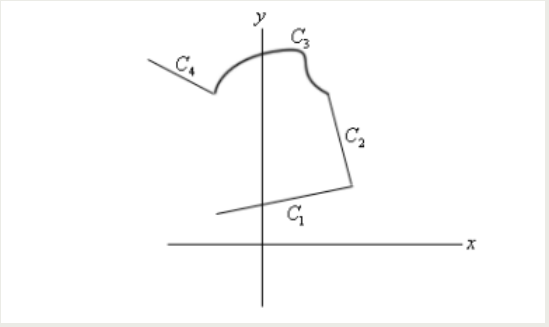
To evaluate a line integral over a piece-wise smooth curve, evaluate the line integral over each of the pieces and then add them up.

The line integral

$$\int_C f(x,y)ds = \int_{C_1} f(x,y)ds + \cdots + \int_{C_n} f(x,y)ds = \sum_{i=1}^{n} \left[\int_{C_1} f(x,y)ds \right]$$

Where $C = C_1 \cup C_2 \cup \cdots \cup C_n = \bigcup_{i=1}^{n} C_i$

Piece-wise smooth curve

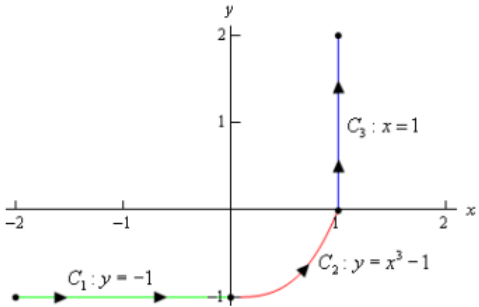


The line integral for some function over the given piecewise curve would be:

$$\int_C f(x,y)ds = \int_{C_1} f(x,y)ds + \int_{C_2} f(x,y)ds + \int_{C_3} f(x,y)ds + \int_{C_4} f(x,y)ds$$

E.g. Evaluate $\int_C 4x^3 ds$, where $C = C_1 \cup C_2 \cup C_3$ is defined as:

$$C_1 : x = t, y = -1, -2 \leq t \leq 0$$
$$C_2 : x = t, y = t^3 - 1, 0 \leq t \leq 1$$
$$C_3 : x = 1, y = t, 1 \leq t \leq 2$$


y
2
1
−2
−1
1
2
x
−1
$C_3 : x = 1$
$C_2 : y = x^3 - 1$
$C_1 : y = -1$

Solution

The line integral over each of these curves are

$$\int_{C_1} 4x^3 = \int_{-2}^{0} 4t^3\sqrt{1^2+0^2}dt = \int_{-2}^{0} 4t^3 dt = t^4|_{-2}^{0} = -16$$

$$\int_{C_2} 4x^3 = \int_{0}^{1} 4t^3\sqrt{1^2+(3t^2)^2}dt = \int_{0}^{1} 4t^3\sqrt{1+9t^4}dt = \frac{2}{27}(1+9t^4)^{\frac{3}{2}}|_{0}^{1} = 2.268$$

$$\int_{C_3} 4x^3 = \int_{1}^{2} 4(1)^3\sqrt{0^2+1^2}dt = \int_{1}^{2} 4dt = 4$$

Hence, the result obtained will be : **-16+2.268+4 =-9.732**.



Trial

2 LINE INTEGRAL

1. Evaluate $\int_C 4x^3 ds$ where C is the line segment from (-2,-1) to (1,2)

2. Evaluate $\int_C xds$ for each of the following curves:

a. $C_1 : y = x^2, -1 \leq x \leq 1$

b. C_2: the line segment from (-1,1) to (1,1)

c. C_3 : the line segment from (1,1) to (-1,1)



Line Integral with Respect to x and y

As with the last section we will start with a two-dimensional curve C with parameterization, $x = x(t), y = y(t), a \leq t \leq b$.

The line integral of f with respect to x is, $\int_C f(x,y)dx = \int_a^b f(x(t),y(t))x'(t)dt$ while the line integral of f with respect to y is, $\int_C f(x,y)dy = \int_a^b f(x(t),y(t))y'(t)dt$

These two integral often appear together and so we have the following shorthand notation for these cases; $\int_C Pdx + Qdy = \int_C P(x,y)dx + \int_C Q(x,y)dy$



Evaluate $\int_C yx^2dx + sin(\pi y)dy$ where C is the line segment from (0,2) to (1,4).

Solution

Here is the parameterization of the curve.

$\vec{r} = (1-t)\langle 0,2\rangle + t\langle 1,4\rangle = \langle t, 2+2t\rangle, 0 \leq t \leq 1$

The line integral is,

$$\int_C yx^2dx + sin(\pi y)dy = \int_C yx^2dx + \int_C sin(\pi y)dy$$

$$= \int_0^1 (2+2t)(t)^2(1)dt + \int_0^1 sin(\pi(2+2t))(2)dt$$

$$= (\tfrac{2}{3}t^3 + \tfrac{1}{2}t^2)|_0^1 + \tfrac{1}{\pi}cos(\pi(2+2t))|_0^1$$

$$= \tfrac{7}{6}.$$

Trial

Evaluate $\int_C yx^2dx + sin(\pi y)dy$ where C is the line segment from (1,4) to (0,2).

NB:
Switching the direction of the curve gives a value with an opposite sign. If C is any curve, then:

$$\int_{-C} f(x,y)dx = -\int_C f(x,y)dx \text{ and}$$

$$\int_{-C} f(x,y)dy = -\int_C f(x,y)dy$$

With the combined forms of these two integrals we get , $\int_{-C} Pdx + Qdy = -\int_{C} Pdx + Qdy$

The integrals over three dimensional curves will be:

$$\int_C f(x,y,z)dx = \int_a^b f(x(t),y(t),z(t))x'(t)dt$$

$$\int_C f(x,y,z)dy = \int_a^b f(x(t),y(t),z(t))y'(t)dt$$

$$\int_C f(x,y,z)dz = \int_a^b f(x(t),y(t),z(t))z'(t)dt$$

Where the curve C is parameterized by x=x(t), y=y(t), z=z(t), $a \leq t \leq b$

As with the two dimensional case, these three often occur together as:

$$\int_C Pdx + Qdy + Rdz = \int_C P(x,y,z)dx + \int_C Q(x,y,z)dy + \int_C R(x,y,z)dz$$

E.g. Evaluate $\int_C ydx + xdy + zdz$ where C is given by x=cos(t), y=sin(t), $z = t^2$, $0 \leq t \leq 2\pi$

Solution

$$\int_C ydx + xdy + zdz = \int_C ydx + \int_C xdy + \int_C zdz =$$

$$\int_0^{2\pi} sint(-sint)dt + \int_0^{2\pi} cost(cost)dt + \int_0^{2\pi} t^2(2t)dt$$

$$= -\int_0^{2\pi} sin^2 tdt + \int_0^{2\pi} cos^2 tdt + \int_0^{2\pi} 2t^3$$

$$(-\tfrac{1}{2}(t - \tfrac{1}{2}sin(2t))) + \tfrac{1}{2}(t + \tfrac{1}{2}sin(2t) + \tfrac{1}{2}t^4)|_0^{2\pi}$$

$$= 8\pi^4$$



Line Integrals of Vector Fields

2 LINE INTEGRAL

In this section we are going to evaluate line integrals of vector fields:
$\vec{F}(x,y,z) = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + R(x,y,z)\vec{k}$
and the three dimensional smooth curve. Given $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}, a \leq t \leq b$
the line integral of $\vec{F}$ along C is $\int_C \vec{F}.d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)).\vec{r}'(t)dt$ where
$\vec{F}(\vec{r}(t)) = \vec{F}(x(t), y(t), z(t))$
Line integral of vector fields can also be written as a line integral with respect to arc length as follows:
$\int_C \vec{F}.d\vec{r} = \int_C \vec{F}.\vec{T}ds$ where $\vec{T}(t)$ is the unit tangent vector given by:
$\vec{T} = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$

Eg. Evaluate $\int_C \vec{F}.d\vec{r}$ where $\vec{F}(x,y,z) = 8x^2yz\vec{i} + 5z\vec{j} - 4xy\vec{k}$ and C is the curve given by $\vec{r}(t) = ti + t^2j + t^3k$, $0 \leq t \leq 1$

Solution

The vector field evaluated along the curve is

$\vec{F}(x,y,z,) = (8t^2)(t^2)(t^3)i + 5t^3j - 4t(t^2)k$

$= 8t^7i + 5t^3j - 4t^3k$ and $\vec{r}'(t) = i + 2tj + 3t^2k$

$\vec{F}(\vec{r}(t)).\vec{r}'(t) = 8t^7 + 10t^4 - 12t^5$

The line integral is then given by $\int_C \vec{F}.d\vec{r} = \int_0^1 8t^7 + 10t^4 - 12t^5 dt$

$(t^8 + 2t^5 - 2t^6)|_0^1$

$= 1$

Trial

Evaluate $\int_C \vec{F}.d\vec{r}$ where $\vec{F}(x, y, z) = xzi - yzk$ and C is the line integral segment from $(-1, 2, 0)$ and $(3, 0, 1)$.



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Fundamental Theorem of Line Integrals

Theorem

Suppose that C is a smooth curve given by $\vec{r}, a \leq t \leq b$. Also, suppose that f is a function whose gradient vector, ∇f, is continuous on C. Then,

$$\int_C \nabla f . d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

Note that $\vec{r}(a)$ represents the initial point on C while $\vec{r}(b)$ represents the final point on C.

Example

Evaluate $\int_C \nabla f . d\vec{r}$ where $f(x, y, z) = cos(\pi x) + sin(\pi y) - xyz$ and C is any path that starts from $(1, \frac{1}{2}, 2)$ and ends at $(2, 1, -1)$.



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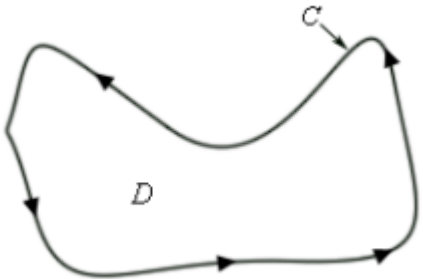
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Green's Theorem

4 GREEN'S THEOREM

This theorem explains the relationship between line integrals on closed path and double integrals.

Below is a simple closed curve C that has a positive orientation. D is the region enclosed by the curve:



The curve is simple and closed because there are no holes in the region D. Also, note that a direction has been put on the curve. We will use the convention here that the curve C has a positive orientation if it is traced out in a counter-clockwise direction.

Given curves/regions such as this we have the following theorem:

$$\int_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$



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Let us define the del ∇ operator as:

$$\nabla = \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k} \tag{1}$$

Whatever function that is listed after the ∇ is substituted into the partial derivatives.
Example

$$\nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k} \tag{2}$$



Using the ∇ operator, we define the **curl** over the vector field $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ as the following cross product:

$$\text{curl}\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \tag{3}$$

$$= (R_y - Q_z)\vec{i} - (R_x - P_z)\vec{j} + (Q_x - P_y)\vec{k} \tag{4}$$



Similarly, **divergence** can be defined in terms of the following dot product.

$$\text{div}\vec{F} = \nabla \bullet \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \tag{5}$$

The following is true about the relationship between curl and divergence;

$$\text{div}(\text{curl}\vec{F}) = 0 \tag{6}$$

Example

Compute $\text{div}\vec{F}$ for $x^2y\vec{i} + xyz\vec{j} - x^2y^2\vec{k}$



Solution

$$\begin{aligned} div\vec{F} &= \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(xyz) + \frac{\partial}{\partial z}(-x^2y^2) \\ &= 2xy + xz \end{aligned}$$

Example

Given the vector field $\vec{F} = yz^2\vec{i} + xy\vec{j} + yz\vec{k}$, verify that $\text{div}(\text{curl}\vec{F}) = 0$.



Solution

$$\begin{aligned}
\text{curl}\vec{F} = \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ yz^2 & xy & yz \end{vmatrix} \\
&= (R_y - Q_z)\vec{i} - (R_x - P_z)\vec{j} + (Q_x - P_y)\vec{k} \\
&= (z - 0)\vec{i} - (0 - 2yz)\vec{j} + (y - z^2)\vec{k} \\
&= z\vec{i} + 2xy\vec{j} + (y - z^2)\vec{k} \qquad (7)
\end{aligned}$$

Now we compute the divergence of equation (7)

$$\begin{aligned} div\vec{F} &= \frac{\partial}{\partial x}(z) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(y - z^2) \\ &= 2z - 2z \\ &= 0 \end{aligned}$$

1. Find *div* **F** for each of the following vector fields:
   - 1.1 $F = \langle xy, yz, xz \rangle$
   - 1.2 $F = \langle yz, xz, xy \rangle$
   - 1.3 $F = \langle xr, yr, zr \rangle$ where $r = \sqrt{x^2 + y^2 + z^2}$
2. Find *curl* **F** for each of the following vector fields:
   - 2.1 $F = \langle xy, yz, xz \rangle$
   - 2.2 $F = \langle yz, xz, xy \rangle$
   - 2.3 $F = \langle xr, yr, zr \rangle$ where $r = \sqrt{x^2 + y^2 + z^2}$



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Thank you for listening!
Any questions?